

TRIANGLE NORTH EXECUTIVE, NC, USA

WMO: 722141

Lat: 36.023N

Lon: 78.330W

Elev: 369

StdP: 14.50

Time Zone: -5.00 (NAE)

Period: 04-19

WBAN: 03731

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	18.8	22.9	0.3	5.6	24.5	4.8	7.1	28.8	17.2	57.8	15.4	53.1	2.5	310	0.366

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.6	94.0	75.2	91.3	74.7	89.9	74.2	78.4	88.2	77.4	87.0	76.4	85.5	4.6	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
76.3	138.9	82.8	74.6	131.3	81.5	73.3	125.3	79.9	42.1	88.8	41.1	86.2	39.9	85.9	84.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.6	12.1	10.6	10.1	97.7	5.2	3.4	6.4	100.1	3.3	102.1	0.4	104.0	-3.4	106.5
			8.6	81.0	4.6	1.9	5.2	82.3	2.5	83.4	-0.1	84.5	-3.4	85.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	60.2	40.8	44.1	50.3	60.0	67.8	75.5	78.6	77.5	71.8	60.8	50.1	44.5
	DBStd	15.57	10.37	9.97	9.91	8.61	7.05	5.48	4.31	4.80	5.55	8.19	8.81	9.46
	HDD50	1021	318	218	125	16	0	0	0	0	0	10	107	227
	HDD65	3336	749	587	464	196	57	3	0	0	12	181	450	637
	CDD50	4759	34	53	134	317	551	764	888	853	653	345	110	57
	CDD65	1600	0	2	8	47	143	316	423	388	214	52	4	3
	CDH74	14279	0	12	86	447	1185	2984	4094	3499	1569	375	20	8
	CDH80	5588	0	0	9	98	350	1225	1816	1466	537	87	0	0
Wind	WSAvg	3.6	4.0	4.5	4.8	4.9	3.6	2.9	2.8	2.6	3.1	3.1	3.5	3.5
Precipitation	PrecAvg	45.4	3.8	3.5	4.1	3.1	4.0	3.7	4.6	5.1	4.1	3.3	3.1	3.1
	PrecMax	65.1	7.8	7.8	8.2	8.3	10.3	10.0	10.2	12.2	20.8	11.5	7.8	7.3
	PrecMin	34.2	0.9	0.7	1.0	0.4	1.5	0.4	1.2	0.5	0.6	0.0	0.7	0.4
	PrecStd	7.8	1.7	1.7	1.7	1.7	1.9	2.3	2.0	2.8	3.3	2.2	1.9	1.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	70.3	75.6	81.3	86.8	90.7	97.1	98.2	97.1	92.6	87.8	76.7	73.2
		MCWB	60.5	62.3	61.9	64.3	71.8	73.7	76.1	77.4	72.2	72.1	63.1	65.0
	2%	DB	66.0	70.4	76.5	82.3	87.6	92.7	94.2	92.9	89.5	81.8	72.8	70.0
		MCWB	59.0	59.3	60.0	64.3	70.7	74.0	76.1	75.7	72.6	69.5	61.8	64.5
	5%	DB	63.0	66.1	72.3	79.4	83.9	90.1	91.2	90.5	86.1	78.2	69.7	64.8
		MCWB	56.3	56.5	58.4	63.1	70.1	73.7	75.6	74.9	72.0	68.1	61.9	60.6
	10%	DB	58.6	62.4	67.9	75.2	81.1	87.5	89.8	88.1	82.3	74.1	64.5	61.1
		MCWB	52.6	54.0	57.5	62.4	68.2	73.0	75.0	74.5	70.8	66.3	58.5	55.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	64.4	65.6	66.1	71.6	75.2	78.4	80.8	80.5	78.0	75.7	68.6	68.6
		MCDB	66.0	71.7	74.5	77.2	83.7	86.9	90.9	91.1	83.2	81.4	70.9	71.5
	2%	WB	60.9	62.6	63.8	68.5	73.1	76.8	78.6	78.4	76.1	73.2	65.9	65.1
		MCDB	64.9	66.9	70.1	75.8	81.7	87.3	89.4	88.0	83.0	77.8	70.0	68.3
	5%	WB	57.2	59.3	62.1	66.1	71.9	75.6	77.4	77.3	74.6	71.5	63.0	62.1
		MCDB	61.7	65.8	68.5	74.8	80.5	86.1	88.1	86.4	81.4	76.4	66.8	64.7
	10%	WB	52.9	54.2	59.4	64.2	70.4	74.4	76.4	76.1	73.2	68.3	59.9	56.8
		MCDB	56.8	60.1	66.4	71.7	77.9	83.7	86.1	84.5	79.2	72.9	64.9	60.8
Mean Daily Temperature Range	5% DB	MDBR	18.9	20.6	22.3	23.2	20.9	20.3	19.6	18.9	18.8	21.2	21.4	19.1
		MCDBR	23.8	24.8	26.2	26.6	23.2	22.5	21.4	21.4	22.9	23.4	24.5	21.5
		MCWBR	17.9	16.2	14.0	11.8	9.7	8.3	7.5	7.4	9.2	12.5	15.6	16.4
	5% WB	MCDBR	21.6	23.5	22.1	20.8	19.5	20.0	19.1	18.5	18.1	18.5	19.6	19.5
		MCWBR	18.7	17.8	14.7	11.1	9.3	8.5	7.8	7.6	8.6	11.1	15.6	16.8
Clear-Sky Solar Irradiance	taub	0.310	0.327	0.357	0.403	0.452	0.498	0.534	0.528	0.416	0.369	0.328	0.312	
	taud	2.504	2.451	2.404	2.275	2.172	2.088	2.005	2.007	2.319	2.431	2.504	2.535	
	Ebn at Noon	282	290	289	279	266	253	243	241	266	270	273	274	
	Edh at Noon	26	30	35	42	47	51	56	54	38	31	26	24	
All-Sky Solar Radiation	RadAvg	805	1026	1349	1707	1891	2032	1921	1735	1436	1160	886	701	
	RadStd	56	118	98	126	146	135	108	91	122	147	71	59	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (60 neighbors)	+0.76	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+275	+191

Nomenclature: See separate page